HEALTHCARE CASE 021

97,731

chest X-rays performed during the trial period

HEALTHCARE · CASE 021 · TRIAL

In an NHS trial, AI triage of chest X-rays sped up reports but not lung cancer diagnosis

In a randomized trial across five NHS trusts in England, an AI tool flagged chest X-rays it judged abnormal so they could be reported first. Reports came back sooner. The trial found no significant difference in how soon patients got a CT scan or a lung cancer diagnosis.

Evidence grade B, good evidence Checked 1 Oct 2026
AAI-2026-021 · version 1 1 public source

1 The short version

Between July 2023 and December 2024, five NHS trusts in England took part in a trial of chest X-rays requested by GPs. Of 97,731 X-rays taken, 93,326 were analysed.

1. On days chosen at random, an AI tool, qXR from Qure.ai, sent a notice to the reporting list for each X-ray it judged abnormal, so it could be reported first. On the other days the AI still marked up the X-rays, but sent no notice. On both kinds of day, staff could still review an X-ray straight away under local procedure.
2. On the AI days, X-rays were reported sooner: a median of 34.1 hours, against 47.0 hours on the other days.
3. The trial found no significant difference in how soon patients got a CT scan or a lung cancer diagnosis. The median time from X-ray to CT scan was 53 days either way.
4. The investigators conclude that X-ray AI should not include this kind of prioritization in this setting.

FACTS CHECKED IN THIS SECTION (15)

1. **F001** The LungIMPACT investigators report that between 17 July 2023 and 31 December 2024, 97,731 chest X-rays were performed across five NHS trusts. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
2. **F003** The LungIMPACT investigators report that after data cleaning, 93,326 chest X-rays from 86,945 patients were included in the analysis. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER

3. **F070** The LungIMPACT investigators report that the LungIMPACT trial was a prospective, multicenter, randomized controlled trial conducted between July 2023 and December 2024 across five NHS Trusts in England. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
4. **F071** The LungIMPACT investigators report that participants in the LungIMPACT trial were aged 18 years or older and were attending for a primary care-requested chest X-ray. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
5. **F113** The LungIMPACT investigators report, in Figure 1 (the CONSORT diagram), that 4,405 of the 97,731 LungIMPACT chest X-rays were excluded: 1,382 for a data compliance issue and 3,023 because the chest X-ray was taken at a weekend. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
6. **F074** The LungIMPACT investigators report that LungIMPACT participants aged 18 years or older, attending for a primary care-requested chest X-ray, were block-randomized by day and site to either immediate AI prioritization of reporting or no AI prioritization. DISPUTED
7. **F031** The LungIMPACT investigators report that one AI product was used in the LungIMPACT trial: qXR, from Qure.ai Technologies. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
8. **F048** The LungIMPACT investigators report that on intervention days of the LungIMPACT trial an active notification was sent to the worklist for any 'qXR-suspected-abnormal' case, allowing prioritization for immediate reporting by radiology. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
9. **F083** The LungIMPACT investigators report that in the LungIMPACT trial the AI was applied in both arms at the time of image acquisition, so that the reporter had access to the AI-marked images at the time of reporting in both arms. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
10. **F084** The LungIMPACT investigators report that immediate review of the chest X-ray was permitted, according to local procedure, in both arms of the LungIMPACT trial. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
11. **F023** The LungIMPACT investigators report, in Table 2 under secondary outcomes, that the median time from chest X-ray to chest X-ray report was 34.1 hours, interquartile range 6.6 to 93.1, over 42,814 chest X-rays on AI-prioritization days and 47.0 hours, interquartile range 15.8 to 99, over 44,078 chest X-rays on days without AI prioritization, with a ratio of geometric means of 0.85, 95 percent confidence interval 0.83 to 0.87, P < 0.001. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
12. **F013** The LungIMPACT investigators report no significant difference in the time from chest X-ray to CT scan according to AI prioritization, with a ratio of geometric means of 0.97, 95 percent confidence interval 0.93 to 1.02, P = 0.31. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
13. **F016** The LungIMPACT investigators report no significant difference in the time from chest X-ray to lung cancer diagnosis between AI-prioritization days and days without AI prioritization, with a ratio of geometric means of 0.98, 95 percent confidence interval 0.83 to 1.16, P = 0.84. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
14. **F009** The LungIMPACT investigators report that the median time from chest X-ray to CT scan was 53 days, interquartile range 17 to 145, on AI-prioritization days and 53 days, interquartile range 19 to 141, on days without AI prioritization. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
15. **F125** The LungIMPACT investigators state, in the abstract of D01, that "Therefore, CXR AI deployments should not include worklist prioritization in this context." SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER

B

EVIDENCE GRADE

Good evidence

A randomized trial, registered before it began and reported in a peer-reviewed journal. That starts at B. It would reach A with a clean match against the trial's registry record (not yet checked) and the paper's internal disagreements resolved, or with a second independent examination.

EXHIBIT 1

The numbers

97,731

chest X-rays performed during the trial period

86,945

patients whose X-rays were analysed

34.1 h

median time to an X-ray report on AI days, against 47.0 hours on the other days

53 days

median time from X-ray to CT scan, with or without AI prioritization

FACTS CHECKED IN THIS SECTION (4)

1. **F001** The LungIMPACT investigators report that between 17 July 2023 and 31 December 2024, 97,731 chest X-rays were performed across five NHS trusts. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
2. **F003** The LungIMPACT investigators report that after data cleaning, 93,326 chest X-rays from 86,945 patients were included in the analysis. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
3. **F023** The LungIMPACT investigators report, in Table 2 under secondary outcomes, that the median time from chest X-ray to chest X-ray report was 34.1 hours, interquartile range 6.6 to 93.1, over 42,814 chest X-rays on AI-prioritization days and 47.0 hours, interquartile range 15.8 to 99, over 44,078 chest X-rays on days without AI prioritization, with a ratio of geometric means of 0.85, 95 percent confidence interval 0.83 to 0.87, P < 0.001. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
4. **F009** The LungIMPACT investigators report that the median time from chest X-ray to CT scan was 53 days, interquartile range 17 to 145, on AI-prioritization days and 53 days, interquartile range 19 to 141, on days without AI prioritization. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER

EXHIBIT 2

How the trial ran

21 Feb 2023
A research ethics committee (Cambridge East) approves the trial

14 DAYS

7 Mar 2023
The trial report gives this as the registration date, 132 days before its first X-ray

132 DAYS

17 Jul 2023
The first chest X-rays in the trial

533 DAYS

31 Dec 2024
The last chest X-rays in the trial

156 DAYS

5 Jun 2025
Follow-up of the trial's patients ends

FACTS CHECKED IN THIS SECTION (5)

1. **F120** The LungIMPACT investigators report that the LungIMPACT trial received favourable ethical approval from the East of England Cambridge East Research Ethics Committee, reference 23/EE/0014, on 21 February 2023. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
2. **F119** The LungIMPACT investigators report, in the Reporting Summary, that the LungIMPACT trial was registered on 7 March 2023. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
3. **F121** The LungIMPACT registration date reported in D01, 7 March 2023, is 132 days before 17 July 2023, the first day of the chest X-ray period reported in D01. SUPPORTED · COMPUTED FROM F119, F001
4. **F001** The LungIMPACT investigators report that between 17 July 2023 and 31 December 2024, 97,731 chest X-rays were performed across five NHS trusts. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
5. **F002** The LungIMPACT investigators report that patients were followed up until 5 June 2025. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER

2 Where the paper disagrees with itself

In a few places the trial report says two different things about the same matter. We show both and don't pick one.

? Did any quarter show a difference?

THE PAPER'S TEXT

No statistically significant difference for either main outcome in any quarter.

ITS EXTENDED DATA TABLE 2

Time to CT scan differed in the third quarter of 2024 (P = 0.03).

? Did any secondary outcome differ?

THE PAPER'S TEXT

No differences in secondary outcomes by AI group.

ITS TABLE 2

Time to an X-ray report, listed as a secondary outcome, was shorter on AI days (P < 0.001).

? What was randomized?

THE METHODS, IN ONE PLACE

Participants were randomized in blocks, by day and site.

THE METHODS, ELSEWHERE

Individual patients were not randomized; their X-rays were.

FACTS CHECKED IN THIS SECTION (6)

1. **F017** The LungIMPACT investigators report that no statistically significant differences were found for either primary outcome, time to CT scan or time to lung cancer diagnosis, at any trial site or in any quarter. DISPUTED
2. **F019** The LungIMPACT investigators report, in Extended Data Table 2, a ratio of geometric means of 1.14, 95 percent confidence interval 1.02 to 1.28, P = 0.03, for time to CT scan in the third quarter of 2024, with N = 2,535. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
3. **F022** The LungIMPACT investigators report that there were no differences in secondary outcomes by AI group. DISPUTED
4. **F023** The LungIMPACT investigators report, in Table 2 under secondary outcomes, that the median time from chest X-ray to chest X-ray report was 34.1 hours, interquartile range 6.6 to 93.1, over 42,814 chest X-rays on AI-prioritization days and 47.0 hours, interquartile range 15.8 to 99, over 44,078 chest X-rays on days without AI prioritization, with a ratio of geometric means of 0.85, 95 percent confidence interval 0.83 to 0.87, P < 0.001. SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
5. **F074** The LungIMPACT investigators report that LungIMPACT participants aged 18 years or older, attending for a primary care-requested chest X-ray, were block-randomized by day and site to either immediate AI prioritization of reporting or no AI prioritization. DISPUTED
6. **F077** The LungIMPACT investigators report that in the LungIMPACT trial individual patients were not randomized; their chest X-rays were. DISPUTED

3 In their own words

"AI prioritization of CXR requested by UK primary care has no significant impact on the lung cancer pathway."

THE INVESTIGATORS

"Therefore, CXR AI deployments should not include worklist prioritization in this context."

THE INVESTIGATORS

"A limitation is the use of a single AI product, which will inevitably have different performance characteristics from others."

THE INVESTIGATORS

FACTS CHECKED IN THIS SECTION (3)

1. **F124** The LungIMPACT investigators state, in the abstract of D01, that "AI prioritization of CXR requested by UK primary care has no significant impact on the lung cancer pathway." SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
2. **F125** The LungIMPACT investigators state, in the abstract of D01, that "Therefore, CXR AI deployments should not include worklist prioritization in this context." SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER

3. **F140** The LungIMPACT investigators state, in the Discussion of D01, that "A limitation is the use of a single AI product, which will inevitably have different performance characteristics from others." SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER

4 Still unknown

? What AI prioritization costs. The paper reports no cost figures for this trial; the investigators say a health economics evaluation will be published separately.

? The paper does not state whether Qure.ai received any part of the trial's grant or was paid for the use of qXR.

? How many of the X-rays the AI flagged were in fact reported straight away. The paper does not report it.

? Harms and survival. The paper reports no count of adverse events or harms, and no survival or mortality outcome.

FACTS CHECKED IN THIS SECTION (6)

1. **F153** D01 reports no cost, cost-effectiveness or other monetary figure for AI prioritization or for the AI software in the LungIMPACT trial. UNKNOWN · RECORD SILENT AS OF 2026-09-30
2. **F160** The LungIMPACT investigators state, in the Discussion and again in the Methods of D01, that a health economics evaluation of the LungIMPACT trial will be published separately ("A detailed health economics evaluation will be published separately"; "The health economic analysis will be reported separately."). SUPPORTED · R1 · 1 ORIGIN, 1 EXAMINER
3. **F043** D01 does not state whether Qure.ai Technologies received any part of SBRI Healthcare grant SBRIC01P3039, whether it was paid for the use of qXR in the LungIMPACT trial, or whether the 'SBRI funding received in collaboration with Qure.ai' in Richard W. Lee's declaration is that grant. UNKNOWN · RECORD SILENT AS OF 2026-09-30
4. **F157** D01 does not report how many chest X-rays flagged by the AI on AI-prioritization days were reported immediately, nor the time to report for flagged chest X-rays separately from all chest X-rays.
   UNKNOWN · RECORD SILENT AS OF 2026-09-30
5. **F154** D01 reports no count of adverse events or patient harms in either arm of the LungIMPACT trial.
   UNKNOWN · RECORD SILENT AS OF 2026-09-30
6. **F156** D01 reports no survival or mortality outcome for the patients in the LungIMPACT trial.
   UNKNOWN · RECORD SILENT AS OF 2026-09-30

HOW TO CITE THIS CASE

Applied AI Case Study Library. “In an NHS trial, AI triage of chest X-rays sped up reports but not lung cancer diagnosis.” Case AAI-2026-021, version 1, checked 1 Oct 2026. https://casestudy.aioperationsmanagement.ai/cases/2026/aai-2026-021-lungimpact-ai-chest-xray-prioritization-rct/

Printed from the live case, where every statement links to its evidence.

IN CONTEXT

THEME

Healthcare

SOURCES READ

▼ **1 document**

1. AI-based chest X-ray prioritization in the lung cancer diagnostic pathway: the LungIMPACT randomized controlled trial

PERIOD COVERED

21 Feb 2023 to 5 Jun 2025

EVIDENCE CUTOFF

22 September 2026

CHECKED

1 Oct 2026, version 1